

1.2 Memory and storage

How to run this unit, lesson by lesson

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Start here

18 lessons. A realistic estimate is **18 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Memory bay ms-101	1	Lesson_Plans/MS_L01_LessonPlan.pdf
2	Swap space ms-102	1	Lesson_Plans/MS_L02_LessonPlan.pdf
3	Storage warehouse ms-103	1	Lesson_Plans/MS_L03_LessonPlan.pdf
4	Top Trumps ms-104	1	Lesson_Plans/MS_L04_LessonPlan.pdf
5	Pick the storage ms-105	1	Lesson_Plans/MS_L05_LessonPlan.pdf
6	Units lab ms-106	1	Lesson_Plans/MS_L06_LessonPlan.pdf
7	Capacity calculator ms-107	1	Lesson_Plans/MS_L07_LessonPlan.pdf
8	Binary workshop ms-108	1	Lesson_Plans/MS_L08_LessonPlan.pdf
9	Assessment 1 ms-109-test	1	Lesson_Plans/MS_L09_TEST_LessonPlan.pdf
10	Shift and hex lab ms-110	1	Lesson_Plans/MS_L10_LessonPlan.pdf
11	Revision HQ 1.2 part 1 ms-111	1	Lesson_Plans/MS_L11_LessonPlan.pdf
12	Character foundry ms-112	1	Lesson_Plans/MS_L12_LessonPlan.pdf
13	Pixel studio ms-113	1	Lesson_Plans/MS_L13_LessonPlan.pdf
14	Sound studio ms-114	1	Lesson_Plans/MS_L14_LessonPlan.pdf
15	Compression works ms-115	1	Lesson_Plans/MS_L15_LessonPlan.pdf
16	Revision HQ 1.2 part 2 ms-116	1	Lesson_Plans/MS_L16_LessonPlan.pdf
17	Assessment 2 ms-117-test	1	Lesson_Plans/MS_L17_TEST_LessonPlan.pdf
18	Binary blitz ms-bonus	1	Lesson_Plans/MS_BONUS_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard ([teacher.html](#)), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.

2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at topics.html with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- no optional challenge activity — ms-101, ms-102, ms-103, ms-104, ms-105, ms-106, ms-107, ms-108, ms-110, ms-111, ms-112, ms-113, ms-114, ms-115, ms-116
- only one activity asks the pupil to produce an answer rather than choose one — ms-105
- no key question — ms-111, ms-116

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Memory bay ms-101

What are the differences between RAM and ROM?

OUTCOMES	understand the need for primary storage; know the difference between RAM and ROM; know the purpose of RAM and ROM
STATIONS AND ACTIVITIES	6 stations, 11 activities, 27 marks (8 guided, 3 independent, 0 challenge)
RESOURCES	MS_L01_MemoryBay_Lesson.pptx, MS_L01_MemoryBay_Worksheet.docx
PLAN	Lesson_Plans/MS_L01_LessonPlan.pdf

2. Swap space ms-102

Why do we need virtual memory, and what can we do to avoid using it?

OUTCOMES	understand the need for virtual memory; understand how virtual memory works
STATIONS AND ACTIVITIES	6 stations, 12 activities, 20 marks (10 guided, 2 independent, 0 challenge)

RESOURCES	MS_L02_SwapSpace_Lesson.pptx, MS_L02_SwapSpace_Worksheet.docx
PLAN	Lesson_Plans/MS_L02_LessonPlan.pdf

3. Storage warehouse ms-103

Why do computers have secondary storage?

OUTCOMES	understand the need for secondary storage; know the common types of storage; know how each type works
STATIONS AND ACTIVITIES	6 stations, 12 activities, 25 marks (10 guided, 2 independent, 0 challenge)
RESOURCES	MS_L03_StorageWarehouse_Lesson.pptx, MS_L03_StorageWarehouse_Worksheet.docx
PLAN	Lesson_Plans/MS_L03_LessonPlan.pdf

4. Top Trumps ms-104

What are the differences between secondary storage devices?

OUTCOMES	know the characteristics of storage devices; be able to compare devices using those characteristics
STATIONS AND ACTIVITIES	6 stations, 12 activities, 24 marks (10 guided, 2 independent, 0 challenge)
RESOURCES	MS_L04_TopTrumpes_Lesson.pptx, MS_L04_TopTrumpes_Worksheet.docx
PLAN	Lesson_Plans/MS_L04_LessonPlan.pdf

5. Pick the storage ms-105

What features of secondary storage make devices suitable for different situations?

OUTCOMES	understand the suitability of storage devices for certain applications; understand the advantages and disadvantages of devices based on their characteristics
STATIONS AND ACTIVITIES	6 stations, 12 activities, 23 marks (11 guided, 1 independent, 0 challenge)
RESOURCES	MS_L05_PickTheStorage_Lesson.pptx, MS_L05_PickTheStorage_Worksheet.docx
PLAN	Lesson_Plans/MS_L05_LessonPlan.pdf

6. Units lab ms-106

Why is data stored in binary?

OUTCOMES	know the units bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte and petabyte; be able to convert between units; understand that data must be converted to binary to be processed
STATIONS AND ACTIVITIES	6 stations, 15 activities, 27 marks (10 guided, 5 independent, 0 challenge)
RESOURCES	MS_L06_UnitsLab_Lesson.pptx, MS_L06_UnitsLab_Worksheet.docx
PLAN	Lesson_Plans/MS_L06_LessonPlan.pdf

7. Capacity calculator ms-107

How do you calculate data capacity?

OUTCOMES	know what data capacity means; understand how to calculate data capacity requirements
STATIONS AND ACTIVITIES	6 stations, 19 activities, 27 marks (15 guided, 4 independent, 0 challenge)
RESOURCES	MS_L07_CapacityCalculator_Lesson.pptx, MS_L07_CapacityCalculator_Worksheet.docx
PLAN	Lesson_Plans/MS_L07_LessonPlan.pdf

8. Binary workshop ms-108

What can happen when you add two binary numbers together?

OUTCOMES	convert positive denary whole numbers (0-255) to 8-bit binary and back; add two 8-bit binary integers; understand the terms most and least significant bit; understand how overflow errors occur
STATIONS AND ACTIVITIES	6 stations, 15 activities, 15 marks (7 guided, 8 independent, 0 challenge)
RESOURCES	MS_L08_BinaryWorkshop_Lesson.pptx, MS_L08_BinaryWorkshop_Worksheet.docx
PLAN	Lesson_Plans/MS_L08_LessonPlan.pdf

9. Assessment 1 ms-109-test

Your first 1.2 test is done on paper. Warm up with Binary blitz, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	MS_L09_Assessment1_Reflection_Worksheet.docx
PLAN	Lesson_Plans/MS_L09_TEST_LessonPlan.pdf

10. Shift and hex lab ms-110

What is the relationship between denary, binary and hexadecimal?

OUTCOMES	know how to perform left and right binary shifts; understand what a binary shift achieves; convert denary to 2-digit hexadecimal and back; convert between binary and hexadecimal
STATIONS AND ACTIVITIES	6 stations, 15 activities, 15 marks (5 guided, 10 independent, 0 challenge)
RESOURCES	MS_L10_ShiftAndHexLab_Lesson.pptx, MS_L10_ShiftAndHexLab_Worksheet.docx
PLAN	Lesson_Plans/MS_L10_LessonPlan.pdf

11. Revision HQ 1.2 part 1 ms-111

Recap lessons 1 to 10: memory, storage, units, capacity, binary, shifts and hexadecimal.

OUTCOMES	know which parts of the topic I am secure on; finish any outstanding tasks from lessons 1 to 10
STATIONS AND ACTIVITIES	6 stations, 21 activities, 21 marks (18 guided, 3 independent, 0 challenge)
RESOURCES	MS_L11_RevisionHQ12Part1_Lesson.pptx, MS_L11_RevisionHQ12Part1_Worksheet.docx
PLAN	Lesson_Plans/MS_L11_LessonPlan.pdf

12. Character foundry ms-112

How does a computer store characters, and what are the implications for the number of bits used?

OUTCOMES	understand that all data, including text, is represented in binary; know what is meant by a character set; understand the relationship between the number of bits and the number of characters; know about ASCII and Unicode
STATIONS AND ACTIVITIES	6 stations, 16 activities, 23 marks (11 guided, 5 independent, 0 challenge)
RESOURCES	MS_L12_CharacterFoundry_Lesson.pptx, MS_L12_CharacterFoundry_Worksheet.docx
PLAN	Lesson_Plans/MS_L12_LessonPlan.pdf

13. Pixel studio ms-113

How does a computer store graphics?

OUTCOMES	understand how an image is a series of pixels represented in binary; know what is meant by metadata; understand the effect of colour depth and resolution on file size
STATIONS AND ACTIVITIES	6 stations, 15 activities, 15 marks (11 guided, 4 independent, 0 challenge)
RESOURCES	MS_L13_PixelStudio_Lesson.pptx, MS_L13_PixelStudio_Worksheet.docx
PLAN	Lesson_Plans/MS_L13_LessonPlan.pdf

14. Sound studio ms-114

How do computers store sound, and what are the implications for sample rate, duration and bit depth?

OUTCOMES	understand how sound is sampled and stored in digital form; understand how sample rate, duration and bit depth affect file size and quality
STATIONS AND ACTIVITIES	6 stations, 15 activities, 15 marks (11 guided, 4 independent, 0 challenge)
RESOURCES	MS_L14_SoundStudio_Lesson.pptx, MS_L14_SoundStudio_Worksheet.docx
PLAN	Lesson_Plans/MS_L14_LessonPlan.pdf

15. Compression works ms-115

What are the effects on a file for each type of compression?

OUTCOMES	know why data is compressed; understand the difference between lossy and lossless compression; know why some data can only be compressed losslessly
STATIONS AND ACTIVITIES	6 stations, 14 activities, 24 marks (11 guided, 3 independent, 0 challenge)
RESOURCES	MS_L15_CompressionWorks_Lesson.pptx, MS_L15_CompressionWorks_Worksheet.docx
PLAN	Lesson_Plans/MS_L15_LessonPlan.pdf

16. Revision HQ 1.2 part 2 ms-116

Recap lessons 12 to 15: character sets, images, sound and compression, ready for the second test.

OUTCOMES	know which parts of data representation I am secure on; finish outstanding tasks from lessons 12 to 15
STATIONS AND ACTIVITIES	6 stations, 20 activities, 27 marks (18 guided, 2 independent, 0 challenge)
RESOURCES	MS_L16_RevisionHQ12Part2_Lesson.pptx, MS_L16_RevisionHQ12Part2_Worksheet.docx
PLAN	Lesson_Plans/MS_L16_LessonPlan.pdf

17. Assessment 2 ms-117-test

Your second 1.2 test is done on paper. Revise with Revision HQ part 2, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	MS_L17_Assessment2_Reflection_Worksheet.docx
PLAN	Lesson_Plans/MS_L17_TEST_LessonPlan.pdf

18. Binary blitz ms-bonus

Two minutes of denary, binary and hexadecimal conversions against the clock, with speed bonuses and streaks. Replay to beat your personal best.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a bonus challenge. It stretches outcomes already taught and is optional.
STATIONS AND ACTIVITIES	0 stations, 1 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	none recorded
PLAN	Lesson_Plans/MS_BONUS_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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